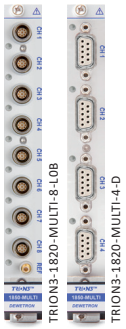
# TRION3-1820-MULTI

DEWETRON

## UNIVERSAL INPUT MODULE




> Sampling: 2 MS/s per channel
> Input types: Voltage, bridge, resistance, RTD, IEPE® current and counter
> Isolation: 350 $V_{DC}$
> CAN: High-speed CAN2.0 port
> Bandwidth: 2 MHz

## SPECIFICATIONS

| TRION3-1820-MULTI | | | |
|---|---|---|---|
| Input channels | | TRION3-1820-MULTI-4-D<br>TRION3-1820-MULTI-8-L0B | 4 channels DSUB connector<br>8 channels 0B LEMO connector |
| Sampling rate / Resolution | High-speed mode<br>Oversampling mode | >1 to 2 MS/s,<br>100 S/s to 1MS/s, | 18-bit; SAR ADC<br>24-bit |
| Analog bandwidth | | 2 MHz | |
| Isolation [1] | | Channel to channel and channel to chassis 350 $V_{DC}$ [1] | |
| Input types | | Ranges | Supported sensors |
| | Voltage | ±0.1 mV to ±100 V[1] freely programmable | - |
| | IEPE® | ±100 mV to ±10 V freely programmable | IEPE® sensors |
| | Bridge | 1 to 1000 mV/V | 4-, 5-, 6-wire full bridge; 3-, 4-, 5-wire ½ bridge;<br>2-, 3-, 4-wire ¼ bridge |
| | Resistance | 10 Ω to 30 kΩ | Potentiometer, Resistance temperature detection:<br>Pt100, Pt200, Pt300, Pt500, Pt2000 (2-, 3-, 4-wire) |
| | Current | ±20 to ±30 mA | 4 to 20 mA sensors;<br>loop powered sensors |
| | MSI | MSI-BR-CH-x: 500 to 50000 pC<br>MSI-BR-TH-x: various TC ranges | Charge output and thermocouple sensors |
| | Counter | Two analog channels can be used as counters; support basic counter functions | |
| | CAN | Channel 1 can be used as CAN port | |
| Voltage input 1 year accuracy | | ±0.02 % of reading ± 0.02 % of range ±20 µV | |
| Linearity | | <50 ppm | |
| Input impedance | 0 to 10 V range<br>10 to 100 V range | 100 MΩ or 1 MΩ (programmable)<br>2 MΩ | |
| Input configuration | | Single ended or differential (programmable) | |
| Input coupling | | DC / AC (high pass filter programmable from 0.16 to 100 Hz) | |
| Over voltage protection | 0 to 10 V range<br>10 to 100 V range | 50 V continuous, 100 V for 60 seconds<br>250 V | |
| Excitation voltage range | | 0 to 24 $V_{DC}$ | |
| Excitation current | | 0.1 to 60 $mA_{DC}$ | |
| Automatic bridge balance | | ±400 % of range | |
| Bridge Completion | | Internal Quarter Bridge Completion: 120 Ω, 350 Ω, 1000 Ω | |
| Bridge features | | freely programmable internal shunt<br>automatic bridge balance<br>automatic lead wire compensation (¼ bridge 3-wire) | |
| Self test (self calibration) | | Each channel is able to perform a complex self test by using internal high precision references | |
| Low pass filter (-3 dB, digital) | Characteristic<br>Filter order | 1 Hz to 40 % of sample rate freely programmable or OFF<br>Bessel or Butterworth<br>$2^{nd}$ , $4^{th}$, $6^{th}$, $8^{th}$ | |
| Analog anti-aliasing filter | | $2^{nd}$ order Bessel,<br>2 MHz, 500 kHz, 100 kHz | |
| Channel to channel phase mismatch | | <60 nsec between channels using the same range | |

[1] for safety reasons it is not allowed to apply more than 47.2 $V_{PEAK}$ or 70 $V_{DC}$